

Electrohydrodynamic Simulation Theory

VSEPR-SIM — Coupled Navier-Stokes / Poisson / Nernst-Planck for Structured Channel Geometries

VSEPR-SIM Project

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1 Scope

This document establishes the theoretical foundation for the `sim/ehd` module of VSEPR-SIM. The module solves a coupled multiphysics problem: incompressible fluid flow, electrostatics, and ionic species transport inside structured (helical-electrode) channel geometries.

The governing equations are:

1. **Continuity and Momentum** (incompressible Navier–Stokes)
2. **Poisson equation** for the electrostatic potential
3. **Nernst–Planck equation** for ionic species transport

2 Governing Equations

2.1 Fluid Flow

Incompressible continuity:

$$\nabla \cdot \mathbf{u} = 0 \quad (1)$$

Momentum conservation with electric body force:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f}_{\text{elec}} \quad (2)$$

where the electric body force is

$$\mathbf{f}_{\text{elec}} = \rho_e \mathbf{E} = \rho_e (-\nabla \phi) \quad (3)$$

and ρ_e is the local space-charge density from ionic species.

Reynolds number for a tube of hydraulic diameter D_h :

$$\text{Re} = \frac{\rho U D_h}{\mu} \quad (4)$$

2.2 Electrostatics

Poisson equation for the electric potential ϕ :

$$\nabla \cdot (\epsilon \nabla \phi) = -\rho_e \quad (5)$$

Electric field:

$$\mathbf{E} = -\nabla \phi \quad (6)$$

Boundary conditions:

$$\phi = V_+ \quad \text{on electrode}_+ \quad (7)$$

$$\phi = V_- \quad \text{on electrode}_- \quad (8)$$

$$\mathbf{n} \cdot (\epsilon \nabla \phi) = 0 \quad \text{on insulating walls} \quad (9)$$

2.3 Ion Transport (Nernst–Planck)

Species conservation for ion i :

$$\frac{\partial c_i}{\partial t} + \nabla \cdot \mathbf{N}_i = 0 \quad (10)$$

Molar flux:

$$\mathbf{N}_i = \underbrace{-D_i \nabla c_i}_{\text{diffusion}} - \underbrace{z_i \mu_i F c_i \nabla \phi}_{\text{electromigration}} + \underbrace{c_i \mathbf{u}}_{\text{convection}} \quad (11)$$

where D_i is diffusivity, z_i is valence, μ_i is ionic mobility, and F is Faraday’s constant. Nernst–Einstein relation (when mobility is not independently known):

$$\mu_i = \frac{|z_i| F D_i}{R T} \quad (12)$$

Space-charge density assembly:

$$\rho_e = F \sum_i z_i c_i \quad (13)$$

3 Coupling Strategy

A segregated outer-iteration scheme is used:

1. Solve flow: $\mathbf{u}, p$ (Navier–Stokes with current $\mathbf{f}_{\text{elec}}$)
2. Solve electric: $\phi, \mathbf{E}$ (Poisson with current ρ_e)
3. Solve species: $c_i, \mathbf{N}_i$ (Nernst–Planck with current $\mathbf{u}, \mathbf{E}$)
4. Update $\rho_e = F \sum z_i c_i$
5. Repeat until convergence

Under-relaxation factors $\alpha_{\text{flow}}, \alpha_{\text{elec}}, \alpha_{\text{species}}$ are applied to stabilise the outer loop.

4 Analytical Baselines

4.1 Hagen–Poiseuille Flow (Straight Tube)

Axial velocity profile:

$$u_z(r) = 2\bar{U} \left(1 - \frac{r^2}{R^2} \right) \quad (14)$$

Pressure drop:

$$\Delta P = \frac{8\mu \bar{U} L}{R^2} \quad (15)$$

4.2 Coaxial Capacitor Field

Potential between inner radius R_{in} and outer radius R_{out} :

$$\phi(r) = V \frac{\ln(R_{\text{out}}/r)}{\ln(R_{\text{out}}/R_{\text{in}})} \quad (16)$$

Radial field:

$$E_r(r) = \frac{V}{r \ln(R_{\text{out}}/R_{\text{in}})} \quad (17)$$

5 Parametric Description

The full parameter set for a helical EHD reactor case is:

$$\mathcal{P} = \{R_{\text{tube}}, L, p_{\text{helix}}, d_{\text{wire}}, n_{\text{turns}}, g, V, Q, \mu, \epsilon, D_i, z_i\} \quad (18)$$

Derived quantities:

$$R_{\text{helix}} = R_{\text{tube}} - g - d_{\text{wire}}/2 \quad (19)$$

$$D_h = 2 R_{\text{tube}} \quad (20)$$

$$\text{Re} = \rho U D_h / \mu \quad (21)$$

6 Engineering Metrics

$$\Delta P = P_{\text{in}} - P_{\text{out}} \quad (22)$$

$$\bar{E} = \frac{1}{V_{\Omega}} \int_{\Omega} |\mathbf{E}| dV \quad (23)$$

$$\dot{N}_i = \int_{A_{\text{out}}} \mathbf{N}_i \cdot \mathbf{n} dA \quad (24)$$

Enhancement factor between geometries:

$$\eta = \frac{\text{performance (modulated geometry)}}{\text{performance (baseline)}} \quad (25)$$

Weighted optimisation objective:

$$J = w_1 \bar{E} + w_2 \dot{N}_{\text{out}} - w_3 \Delta P \quad (26)$$

7 Implementation Mapping

Table 1: EHD module stage structure

Stage	Namespace	Key Classes
1. CAD	<code>vsepr::ehd::cad</code>	HelixGenerator, TubeBodyGenerator, ElectrodeLayout, StepManifest
2. Domain	<code>vsepr::ehd::domain</code>	DomainExtract, NamedRegionRegistry, BCTemplate
3. Physics	<code>vsepr::ehd::physics</code>	FlowModel, ElectrostaticModel, IonTransportModel, CoupledSolver
4. Mesh	<code>vsepr::ehd::mesh</code>	MeshControlSpec, InflationSpec, RefinementZone
5. Post	<code>vsepr::ehd::post</code>	ContourExport, StreamlineExport, SectionProbe, ComparisonTable

8 Phase 1 — Parametric Geometry and Helical Electrode Theory

The CAD stage generates the physical geometry of the helical-electrode EHD reactor from a minimal parameter set $\mathcal{P}$. Every geometric body is derived deterministically; no stochastic elements are introduced.

8.1 Helix Centreline

The electrode wire follows a circular helix of radius R_h and pitch p wound along the tube axis $\hat{z}$. The centreline is parameterised by the azimuthal angle θ :

$$\mathbf{r}_h(\theta) = (R_h \cos \theta, R_h \sin \theta, z_0 + p\theta/(2\pi)) \quad (27)$$

where z_0 is the axial offset of the first turn (the inlet development length) and the total number of discrete points for n_t turns at angular resolution $\Delta\theta$ is

$$N_{\text{pts}} = n_t \cdot (2\pi/\Delta\theta) + 1 \quad (28)$$

The helix radius is derived from the run-card geometry:

$$R_h = R_{\text{tube}} - g - d_w/2 \quad (29)$$

where g is the wall-clearance gap and d_w is the wire diameter.

8.2 Wire Cross-Section

At each centreline station the wire occupies a disk of radius $r_w = d_w/2$ in a plane normal to the local tangent. The tangent vector is

$$\mathbf{t}(\theta) = \frac{d\mathbf{r}_h}{d\theta} \bigg/ \left| \frac{d\mathbf{r}_h}{d\theta} \right|, \quad \frac{d\mathbf{r}_h}{d\theta} = (-R_h \sin \theta, R_h \cos \theta, p/(2\pi)) \quad (30)$$

A local orthonormal frame $(\mathbf{t}, \mathbf{n}, \mathbf{b})$ is constructed via a reference-vector projection. The cross-section circle is then

$$\mathbf{r}_{\text{wire}}(\alpha) = \mathbf{r}_h + r_w(\mathbf{n} \cos \alpha + \mathbf{b} \sin \alpha), \quad \alpha \in [0, 2\pi) \quad (31)$$

8.3 Tube Profile — Straight and Modulated

The inner wall of the tube is a surface of revolution. For the straight geometry the radius is constant:

$$R_{\text{inner}}(z) = R_{\text{tube}} \quad (32)$$

For the modulated (sinusoidal bulge) geometry:

$$R_{\text{inner}}(z) = R_{\text{tube}} + A_m \sin\left(\frac{2\pi z}{\lambda_m}\right) \quad (33)$$

where A_m is the modulation half-amplitude and λ_m is the spatial wavelength. The outer wall is offset by the wall thickness t_w :

$$R_{\text{outer}}(z) = R_{\text{inner}}(z) + t_w \quad (34)$$

Annular ring tessellation at axial station z produces a vertex set

$$\mathbf{r}_{\text{ring}}(\theta_j, z) = (R \cos \theta_j, R \sin \theta_j, z), \quad \theta_j = \frac{2\pi j}{N_\theta} \quad (35)$$

for $j = 0, \dots, N_\theta - 1$.

8.4 Electrode Layout and STEP Manifest

Each electrode is described by a polarity $\sigma \in \{+, -, \text{gnd}\}$, an applied voltage, and a geometry type (helical sweep, coaxial cylinder, or planar pair). The layout assembles all electrodes into a device description from which a STEP-style manifest of named bodies is generated.

The total wire length for a helical electrode is

$$L_{\text{wire}} = n_t \sqrt{(2\pi R_h)^2 + p^2} \quad (36)$$

and the approximate wire volume (used for domain subtraction) is

$$V_{\text{wire}} = \pi r_w^2 L_{\text{wire}} \quad (37)$$

9 Phase 2 — Domain Decomposition and Boundary Classification

The computational domain is partitioned into disjoint material regions:

$$\Omega = \Omega_f \cup \Omega_{s,c} \cup \Omega_{s,d} \quad (38)$$

where Ω_f is the fluid region, $\Omega_{s,c}$ the solid conductors (electrodes), and $\Omega_{s,d}$ the solid dielectrics (tube wall, support structures).

9.1 Fluid Volume Extraction

The net fluid volume is the cylindrical envelope minus all embedded solid volumes:

$$|\Omega_f| = \pi R_{\text{tube}}^2 L_{\text{total}} - \sum_k V_{s,k} \quad (39)$$

where $L_{\text{total}} = L_{\text{inlet}} + L + L_{\text{outlet}}$ and $V_{s,k}$ is the volume of the k -th embedded solid.

9.2 Boundary Surface Taxonomy

Every surface of $\partial\Omega_f$ is assigned a unique tag from a fixed taxonomy:

$$\Gamma_{\text{in}} \quad \text{— inlet face} \quad (40)$$

$$\Gamma_{\text{out}} \quad \text{— outlet face} \quad (41)$$

$$\Gamma_{\text{wall}} \quad \text{— no-slip insulating wall} \quad (42)$$

$$\Gamma_{\text{elec}+} \quad \text{— positive electrode surface} \quad (43)$$

$$\Gamma_{\text{elec}-} \quad \text{— negative / grounded electrode surface} \quad (44)$$

Each tag maps to a triple of physics-specific boundary conditions (flow BC, electric BC, transport BC) as follows:

Table 2: Boundary condition mapping per surface tag			
Surface	Flow	Electric	Transport
Γ_{in}	Velocity inlet $\bar{U}$	Neumann $\mathbf{n} \cdot \nabla \phi = 0$	Fixed $c_{i,0}$
Γ_{out}	Pressure outlet P_0	Neumann insulating	Convective outflow
Γ_{wall}	No-slip $\mathbf{u} = \mathbf{0}$	Neumann insulating	Zero flux $\mathbf{N}_i \cdot \mathbf{n} = 0$
$\Gamma_{\text{elec}+}$	No-slip	Dirichlet $\phi = V_+$	Zero flux
$\Gamma_{\text{elec}-}$	No-slip	Dirichlet $\phi = V_-$	Zero flux

9.3 Multi-Species Boundary Conditions

For N_s ionic species, the inlet boundary assigns a fixed concentration for each species:

$$c_i|_{\Gamma_{\text{in}}} = c_{i,0}, \quad i = 1, \dots, N_s \quad (45)$$

All insulating walls and electrode surfaces enforce zero normal flux:

$$\mathbf{N}_i \cdot \mathbf{n}|_{\Gamma_{\text{wall}} \cup \Gamma_{\text{elec}}} = 0 \quad (46)$$

10 Phase 3 — Coupled Multiphysics: Discretisation and Solution Strategy

This section extends the governing equations from §2 with the discrete field representations, the finite-difference operators, and the segregated coupling algorithm implemented in the `CoupledSolver`.

10.1 Axisymmetric Structured Grid

All three field variables are stored on a collocated (r, z) grid with N_r radial and N_z axial cells:

$$r_i = \left(i + \frac{1}{2}\right) \Delta r, \quad z_j = \left(j + \frac{1}{2}\right) \Delta z \quad (47)$$

where $\Delta r = R_{\text{tube}}/N_r$ and $\Delta z = L_{\text{total}}/N_z$. The total cell count is $N_{\text{cells}} = N_r \times N_z$.

10.2 Flow Field Storage

Each flow cell stores velocity $\mathbf{u}_{ij} = (u_r, u_\theta, u_z)$ and pressure p_{ij} . The flow field is initialised to the Hagen–Poiseuille baseline:

$$u_z(r_i) = 2\bar{U} \left(1 - \frac{r_i^2}{R_{\text{tube}}^2}\right), \quad p(z_j) = \Delta P_{\text{HP}} \left(1 - \frac{j}{N_z}\right) \quad (48)$$

where $\Delta P_{\text{HP}} = 8\mu_f \bar{U} L_{\text{total}}/R_{\text{tube}}^2$.

The maximum velocity magnitude over the grid is

$$u_{\text{max}} = \max_{i,j} |\mathbf{u}_{ij}| \quad (49)$$

10.3 Electric Field Storage and Finite-Difference Gradient

Each electric cell stores potential ϕ_{ij} , field $\mathbf{E}_{ij}$, and local charge density $\rho_{e,ij}$.

The field is initialised to the coaxial analytical solution (§10, equations (27)–(37) are in Phase 1). After each outer iteration the gradient is recomputed via second-order central differences:

$$E_{r,ij} = -\frac{\phi_{i+1,j} - \phi_{i-1,j}}{2\Delta r}, \quad E_{z,ij} = -\frac{\phi_{i,j+1} - \phi_{i,j-1}}{2\Delta z} \quad (50)$$

Interior cells only ($1 \leq i \leq N_r - 2$, $1 \leq j \leq N_z - 2$).

Peak and volume-averaged field magnitudes:

$$E_{\text{max}} = \max_{i,j} |\mathbf{E}_{ij}| \quad (51)$$

$$\bar{E} = \frac{1}{|\Omega|} \sum_{i,j} |\mathbf{E}_{ij}| \Delta V_{ij} \approx \frac{1}{N_{\text{cells}}} \sum_{i,j} |\mathbf{E}_{ij}| \quad (52)$$

10.4 Species Field and Flux Discretisation

For each species i the concentration gradient at cell (i_r, i_z) is

$$\nabla c_i|_{i_r, i_z} \approx \left(\frac{c_{i_r+1, i_z} - c_{i_r-1, i_z}}{2 \Delta r}, 0, \frac{c_{i_r, i_z+1} - c_{i_r, i_z-1}}{2 \Delta z} \right) \quad (53)$$

The discrete Nernst–Planck flux (equation (11)) at a cell is then

$$\mathbf{N}_i^{(k)} = -D_i \nabla c_i - z_i \mu_i F c_i \nabla \phi + c_i \mathbf{u} \quad (54)$$

where all fields are evaluated at the (k) -th outer iteration.

10.5 Charge-Density Assembly

After species fluxes are updated, the charge density at each cell is

$$\rho_{e, ij} = F \sum_{s=1}^{N_s} z_s c_{s, ij} \quad (55)$$

This feeds back into the Poisson solve for the next outer iteration.

10.6 Segregated Outer-Iteration Algorithm

The coupling loop (Algorithm 10.6) is:

1. **Flow sub-solve:** update $\mathbf{u}, p$ from Navier–Stokes with current $\rho_e \mathbf{E}$ body force. (Production: SIMPLE or projection method; baseline: retain Poiseuille.)
2. **Electrostatic sub-solve:** recompute $\mathbf{E}$ from ϕ via (50). (Production: assemble ρ_e into Poisson, solve with CG/GMRES.)
3. **Transport sub-solve:** evaluate $\mathbf{N}_i$ for all species using current $\mathbf{u}$ and $\mathbf{E}$. (Production: upwind advection + implicit diffusion.)
4. **Charge update:** assemble ρ_e from (55).
5. **Convergence check:** compute residual

$$\mathcal{R} = \sqrt{\frac{1}{N_{\text{cells}}} \sum_{i,j} \rho_{e, ij}^2} \quad (56)$$

Converged if $\mathcal{R} < \varepsilon_{\text{tol}}$.

Under-relaxation is applied to each sub-solve:

$$\xi^{(k+1)} = \alpha \xi_{\text{new}} + (1 - \alpha) \xi^{(k)} \quad (57)$$

with factors α_{flow} , α_{elec} , α_{species} specified in the run card.

10.7 Outlet Molar Flux Integration

The species molar flow rate across the outlet is computed as

$$\dot{N}_i = \int_{\Gamma_{\text{out}}} \mathbf{N}_i \cdot \mathbf{n} dA \approx \sum_{i_r=0}^{N_r-1} N_{i,z}|_{i_r, N_z-1} \cdot 2\pi r_{i_r} \Delta r \quad (58)$$

10.8 Accumulation Index

A measure of species focusing or depletion by the electric field:

$$\mathcal{A}_i = \frac{\max_{i_r, i_z} c_i}{\min_{i_r, i_z} c_i} \quad (59)$$

$\mathcal{A}_i = 1$ for a perfectly uniform field; large values indicate strong electromigration-driven concentration.

11 Phase 4 — Mesh Sizing and Adaptive Refinement

Although the current solver uses a uniform structured grid, the mesh-control framework specifies element-size fields that external meshers (gmsh, ANSYS Meshing, snappyHexMesh) consume for unstructured or hybrid mesh generation.

11.1 Global and Local Sizing

A base element size h_0 is set proportional to the tube radius:

$$h_0 = \alpha_h R_{\text{tube}} \quad (60)$$

with $\alpha_h \approx 0.12$. Local sizes are imposed near critical features:

$$h_{\text{elec}} = \beta_e d_w \quad (\beta_e \approx 0.06) \quad (61)$$

$$h_{\text{wall}} = \beta_w R_{\text{tube}} \quad (\beta_w \approx 0.025) \quad (62)$$

A geometric growth rate γ transitions from the fine local size to the global base size:

$$h_{n+1} = \gamma h_n, \quad \gamma \approx 1.2 \quad (63)$$

11.2 Boundary-Layer Inflation

Near viscous walls and electrode surfaces, prism (inflation) layers resolve the velocity and concentration boundary layers. The first cell height is

$$\delta_1 = \alpha_\delta R_{\text{tube}} \quad (64)$$

with $\alpha_\delta \approx 0.003$. Subsequent layers grow geometrically:

$$\delta_k = \delta_1 \gamma_{\text{inf}}^{k-1}, \quad k = 1, \dots, N_{\text{layers}} \quad (65)$$

Total inflation thickness must not exceed δ_{max} .

11.3 Refinement Zones — Electrode Vicinity

For each of the n_t helix turns, a cylindrical refinement zone is centred on the helix wire:

$$\mathcal{Z}_t = \{(r, z) : |r - R_h| \leq d_w, |z - z_{c,t}| \leq 0.6p\} \quad (66)$$

where $z_{c,t} = z_0 + (t + \frac{1}{2})p$ is the axial centre of turn t .

The total number of refinement zones scales linearly with turn count:

$$N_{\text{zones}} = n_t + N_{\text{transition}} \quad (67)$$

where $N_{\text{transition}}$ accounts for inlet/outlet transition zones.

12 Phase 5 — Postprocessing, Probes, and Comparison Framework

12.1 Contour Field Export

Scalar field quantities $\psi(r_i, z_j)$ are exported as structured (r, z, ψ) CSV grids. The supported quantities and their definitions are:

$$|\mathbf{u}|_{ij} = \sqrt{u_{r,ij}^2 + u_{\theta,ij}^2 + u_{z,ij}^2} \quad (\text{velocity magnitude}) \quad (68)$$

$$p_{ij} \quad (\text{pressure}) \quad (69)$$

$$\phi_{ij} \quad (\text{electric potential}) \quad (70)$$

$$|\mathbf{E}|_{ij} = \sqrt{E_{r,ij}^2 + E_{z,ij}^2} \quad (\text{field magnitude}) \quad (71)$$

$$\rho_{e,ij} \quad (\text{charge density}) \quad (72)$$

$$c_{s,ij} \quad (\text{species concentration}) \quad (73)$$

12.2 Streamline Tracing

Streamlines are integrated through the velocity field by forward-Euler advection from inlet seed positions:

$$\mathbf{x}^{(n+1)} = \mathbf{x}^{(n)} + \Delta s \frac{\mathbf{u}(\mathbf{x}^{(n)})}{|\mathbf{u}(\mathbf{x}^{(n)})|} \quad (74)$$

with step size $\Delta s = f_s \min(\Delta r, \Delta z)$ and step fraction $f_s \approx 0.3$. Seeds are distributed uniformly across the inlet radius:

$$r_{0,k} = \frac{R_{\text{tube}}(k + \frac{1}{2})}{N_{\text{seeds}}}, \quad k = 0, \dots, N_{\text{seeds}} - 1 \quad (75)$$

12.3 Section Probes — Radial and Axial

A *radial probe* extracts the profile $\psi(r)$ at a fixed axial station j^* :

$$\{(r_i, \psi_{ij^*})\}_{i=0}^{N_r-1} \quad (76)$$

An *axial probe* extracts $\psi(z)$ at a fixed radial index i^* :

$$\{(z_j, \psi_{i^*j})\}_{j=0}^{N_z-1} \quad (77)$$

Both probes support all six field quantities listed in §12.

12.4 Comparison Tables and Enhancement Factor

Given two solver metric sets $\mathcal{M}_{\text{base}}$ (straight) and $\mathcal{M}_{\text{mod}}$ (modulated), the enhancement factor for each metric is

$$\eta_m = \frac{m_{\text{mod}}}{m_{\text{base}}}, \quad m \in \{\Delta P, E_{\text{max}}, \bar{E}, u_{\text{max}}, \text{Re}, \dot{N}_i, \mathcal{A}_i\} \quad (78)$$

12.5 Multi-Objective Optimisation

The weighted objective function for geometry optimisation is

$$J(\mathcal{P}) = w_1 \bar{E} + w_2 \dot{N}_{\text{out}} - w_3 \Delta P \quad (79)$$

The negative sign on ΔP penalises excessive pressure loss. Pareto exploration sweeps weight vectors $\mathbf{w} = (w_1, w_2, w_3)$ to identify non-dominated geometry candidates.

13 Phase 6 — Dimensional Analysis, Sensitivity, and Verification

13.1 Dimensionless Groups

The EHD system is governed by the following key dimensionless numbers:

$$\text{Re} = \frac{\rho \bar{U} D_h}{\mu_f} \quad (\text{Reynolds — inertia/viscous}) \quad (80)$$

$$\text{Eh} = \frac{\epsilon \Delta V}{\mu_f \bar{U} D_h} \quad (\text{electric Hartmann analogue}) \quad (81)$$

$$\text{Pe}_i = \frac{\bar{U} D_h}{D_i} \quad (\text{Péclet — convection/diffusion}) \quad (82)$$

$$\text{Da}_i = \frac{|z_i| \mu_i F \Delta V}{D_i} \quad (\text{Damköhler-migration — migration/diffusion}) \quad (83)$$

These groups classify the dominant physics regime: for $\text{Re} \ll 2300$ the flow is laminar; for $\text{Pe}_i \gg 1$ convection dominates species transport; for $\text{Da}_i \gg 1$ electromigration dominates diffusion.

13.2 Parametric Sensitivity — Voltage Sweep

At constant flow rate, increasing ΔV scales the electric field linearly:

$$E_{\text{max}}(\Delta V) = \frac{\Delta V}{R_{\text{in}} \ln(R_{\text{out}}/R_{\text{in}})} \propto \Delta V \quad (84)$$

while the pressure drop remains approximately constant (the Lorentz body force is small compared to the viscous force for typical EHD operating conditions at low Re).

13.3 Parametric Sensitivity — Velocity Sweep

The analytical pressure drop scales linearly with mean velocity:

$$\Delta P = \frac{8 \mu_f \bar{U} L_{\text{total}}}{R_{\text{tube}}^2} \propto \bar{U} \quad (85)$$

and the Reynolds number also scales linearly:

$$\text{Re} = \frac{\rho \bar{U} D_h}{\mu_f} \propto \bar{U} \quad (86)$$

13.4 Memory Footprint Estimation

Per-cell storage requirements:

$$S_{\text{flow}} = |\text{FlowCell}| = 32 \text{ bytes} \quad (3 \times 8 \text{ B velocity} + 8 \text{ B pressure}) \quad (87)$$

$$S_{\text{elec}} = |\text{ElectricCell}| = 40 \text{ bytes} \quad (8 \text{ B } \phi + 3 \times 8 \text{ B } \mathbf{E} + 8 \text{ B } \rho_e) \quad (88)$$

$$S_{\text{spec}} = |\text{SpeciesCell}| = 32 \text{ bytes} \quad (8 \text{ B } c_i + 3 \times 8 \text{ B } \mathbf{N}_i) \quad (89)$$

Total memory for N_s species on an $N_r \times N_z$ grid:

$$M_{\text{total}} = N_r N_z (S_{\text{flow}} + S_{\text{elec}} + N_s S_{\text{spec}}) \quad (90)$$

For the default single-species case ($N_s = 1$) on a 160×800 grid (128,000 cells):

$$M \approx 128,000 \times (32 + 40 + 32) = 13.31 \text{ MB} \quad (91)$$

13.5 Determinism Verification

A core design requirement of the VSEPR-SIM platform is bit-exact reproducibility. All field computations are IEEE 754 double-precision with deterministic reduction order. Verification confirms that repeated execution produces identical output metrics:

$$\left| \Delta P^{(a)} - \Delta P^{(b)} \right| = 0, \quad \left| E_{\max}^{(a)} - E_{\max}^{(b)} \right| = 0, \quad \left| u_{\max}^{(a)} - u_{\max}^{(b)} \right| = 0 \quad (92)$$

for any two invocations (a) , (b) with identical input parameters.

14 Phase 7 — EHD Pump Configurations and Pumping Mechanisms

The preceding phases treated a single device topology (helical wire electrode inside a cylindrical tube). Real EHD pump systems span a broader design space with at least four principal electrode configurations and three distinct pumping mechanisms. This phase formalises the theoretical basis for each.

14.1 Four Principal Configurations

14.1.1 Configuration (a): Planar Electrode Channel

Two parallel planar electrodes separated by a gap H drive flow through a rectangular fluidic channel of width W and streamwise length L_{ch} .

Uniform field. The electric field between infinite parallel plates is

$$E_y = \frac{\Delta V}{H} \quad (93)$$

and the potential varies linearly:

$$\phi(y) = \Delta V \left(1 - \frac{y}{H} \right) \quad (94)$$

Plane-Poiseuille baseline. The velocity profile between no-slip plates is

$$u_z(y) = \frac{6 \bar{U}}{H^2} y (H - y) \quad (95)$$

with peak velocity $u_{\max} = \frac{3}{2} \bar{U}$ at the midplane. The associated pressure drop is

$$\Delta P = \frac{12 \mu_f \bar{U} L_{ch}}{H^2} \quad (96)$$

Hydraulic diameter.

$$D_h = \frac{2 H W}{H + W} \quad (97)$$

14.1.2 Configuration (b): Needle-Ring Ion Injection

A sharp needle electrode (tip curvature r_{tip}) faces a grounded ring electrode (inner radius R_{ring}) across an axial gap d_{gap} . Corona discharge at the needle tip injects ions that are accelerated by the field, transferring momentum to the neutral gas/liquid.

Tip field enhancement. For a hyperboloidal needle the field at the tip is

$$E_{\text{tip}} \approx \frac{\Delta V}{r_{\text{tip}} \ln(2 d_{\text{gap}}/r_{\text{tip}})} \quad (98)$$

The ratio $E_{\text{tip}}/(\Delta V/d_{\text{gap}})$ is the geometric enhancement factor, typically $\gg 1$ for sharp tips.

Peek's corona onset criterion. In air at relative density δ (STP $\delta = 1$) the onset field for positive corona is

$$E_{\text{onset}} = 3.1 \times 10^6 \delta \left(1 + \frac{0.308}{\sqrt{\delta} r_{\text{tip}}} \right) \quad [\text{V/m}] \quad (99)$$

Ion-drag pressure. The generated pressure head from corona current I is

$$\Delta P_{\text{ion}} \approx \frac{I d_{\text{gap}}}{\mu_i A_{\text{ring}}} \quad (100)$$

where μ_i is the ion mobility and $A_{\text{ring}} = \pi R_{\text{ring}}^2$.

14.1.3 Configuration (c): Stacked Perforated Disk Electrodes

N_d circular disks of radius R_{disk} and thickness t_d are arranged along the flow axis with alternating polarity. Each disk is perforated with N_p through-holes of radius r_p .

Stack geometry. Total stack length:

$$L_{\text{stack}} = N_d t_d + (N_d - 1) d_s \quad (101)$$

where d_s is the inter-disk spacing. The open-area fraction is

$$\alpha_{\text{OA}} = \frac{N_p \pi r_p^2}{\pi R_{\text{disk}}^2} = \frac{N_p r_p^2}{R_{\text{disk}}^2} \quad (102)$$

Inter-disk field. Between adjacent disks (uniform approximation):

$$E_{\text{gap}} = \frac{\Delta V}{d_s} \quad (103)$$

Flow character. The device produces both axial through-flow (via perforations) and rotational swirl in the inter-disk gaps. The effective cross-section for through-flow is $A_{\text{eff}} = N_p \pi r_p^2$.

14.1.4 Configuration (d): Triangular Prism Slit-Electrode

An array of N_{prism} triangular prism electrodes with base b and height h_p are separated by slit gaps of width w_s . The prism tips produce strongly non-uniform fields that drive cross-flow patterns.

Array pitch.

$$p_{\text{array}} = b + w_s \quad (104)$$

Total array width: $W_{\text{array}} = N_{\text{prism}} b + (N_{\text{prism}} - 1) w_s$.

Tip field enhancement. For an isosceles triangle with half-angle $\alpha = \arctan(b/2h_p)$, the field at the apex is enhanced relative to the uniform slit field:

$$E_{\text{tip}} \approx \frac{\Delta V}{w_s} \cdot \frac{1}{\sin \alpha} \quad (105)$$

The factor $\beta = 1/\sin \alpha$ captures the lightning-rod concentration of field lines at sharp tips.

14.2 Three EHD Pumping Mechanisms

14.2.1 Mechanism 1: Coulomb / Ion-Drag Force

The dominant mechanism in ion-injection and corona-discharge devices. Space charge ρ_e interacts with the local field:

$$\mathbf{f}_C = \rho_e \mathbf{E} \quad (106)$$

This produces a volumetric body force that is injected into the momentum equation. The characteristic Coulomb pressure scale is

$$P_C \sim \epsilon E^2 = \epsilon \left(\frac{\Delta V}{d} \right)^2 \quad (107)$$

14.2.2 Mechanism 2: Dielectrophoretic (DEP) Force

Arises from permittivity gradients in non-uniform electric fields. The time-averaged DEP force on a polarisable medium is

$$\mathbf{f}_{\text{DEP}} = \frac{1}{2} \epsilon_0 K_{\text{CM}} \nabla |\mathbf{E}|^2 \quad (108)$$

where $K_{\text{CM}} = \text{Re}[K(\omega)]$ is the Clausius-Mossotti factor (bounded between -0.5 and $+1.0$ for spherical particles).

The characteristic DEP velocity is

$$u_{\text{DEP}} \sim \frac{\epsilon K_{\text{CM}} E^2 d}{\mu_f} \quad (109)$$

14.2.3 Mechanism 3: Electroosmotic Flow (EOF)

In micro-channels with charged walls, the electric double layer (EDL) interacts with the tangential field to produce a plug-flow slip at the wall.

Helmholtz-Smoluchowski slip.

$$\mathbf{u}_{\text{slip}} = -\frac{\epsilon \zeta}{\mu_f} \mathbf{E}_t \quad (110)$$

where ζ is the wall zeta potential and $\mathbf{E}_t$ is the tangential (wall-parallel) component of the field.

Debye length. The characteristic thickness of the EDL is

$$\lambda_D = \sqrt{\frac{\epsilon R T}{2 F^2 c_0}} \quad (111)$$

The thin-EDL approximation ($\kappa H \gg 1$ where $\kappa = 1/\lambda_D$) is valid when the Debye length is much smaller than the channel dimension.

EOF flow rate (capillary). In a cylindrical capillary of radius R_c :

$$Q_{\text{EOF}} = -\frac{\epsilon \zeta E_z \pi R_c^2}{\mu_f} \quad (112)$$

EOF pressure head. Under closed-channel conditions the electroosmotic flow generates a back-pressure:

$$\Delta P_{\text{EOF,cyl}} = \frac{8 \epsilon |\zeta| |E_z| L}{R_c^2} \quad (113)$$

$$\Delta P_{\text{EOF,planar}} = \frac{12 \epsilon |\zeta| |E_z| L}{H^2} \quad (114)$$

14.3 Topology–Mechanism Affinity

Not every mechanism is equally relevant to every configuration. The natural pairings are:

Table 3: Primary pumping mechanisms by device topology

Configuration	Coulomb	DEP	EOF	Typical Application
(a) Planar channel	✓	—	✓	Micro-pump, lab-on-chip
(b) Needle-ring	✓	—	—	Gas pump, ion wind
(c) Disk stack	✓	—	—	Multi-stage liquid pump
(d) Prism slit	✓	✓	—	Particle manipulation
Helical wire (orig.)	✓	—	—	Tubular reactor

15 Phase 8: Metallic Particle Combustion and Reactive Multiphase Flow

This phase introduces a Lagrangian–Eulerian reactive multiphase model coupling discrete burning particles with the carrier-gas continuum. Five representative fuel types—methane (CH_4), iron (Fe), aluminum (Al), boron/aluminum (B/Al), and zirconium (Zr)—are modelled following the experimental observations of Bergthorson (2015) and the classical combustion theory of Glassman & Yetter.

15.1 Fuel Classification

The five fuels span three combustion regimes:

Table 4: Fuel types, combustion regimes, and characteristic emission species

Fuel	Formula	Regime	Emitter	T_{ad} (K)
Methane	CH_4	Premixed gas	CH^*	2 223
Iron	Fe	Surface	FeO	2 340
Aluminum	Al	Vapour-phase	AlO	3 673
Boron/Aluminum	B/Al	Hybrid	BO_2	3 450
Zirconium	Zr	Hybrid	ZrO	2 950

15.2 d^n -Law Burning Model

The classical shrinking-particle model generalises the d^2 -law to an arbitrary exponent n :

$$d(t)^n = d_0^n - K t \quad (115)$$

where K (m^n/s) is the *burning-rate constant*. For vapour-phase combustion (Al, B/Al) the classical $n = 2$ applies; for surface-limited heterogeneous oxidation (Fe) $n \approx 1$; and for hybrid fuels (Zr) $n \approx 1.8$.

The particle burnout time follows immediately:

$$t_{\text{burn}} = \frac{d_0^n}{K} \quad (116)$$

The instantaneous mass-loss rate for a spherical particle of density ρ_p is:

$$\frac{dm}{dt} = -\frac{\pi}{2} \rho_p K d^{2-n} \quad (117)$$

which is *constant* for $n = 2$ (classical d^2 -law) and linearly proportional to d for $n = 1$ (surface regime).

15.3 Burning-Rate Corrections

Temperature correction (Arrhenius).

$$K(T) = K_0 \exp \left[\frac{E_a}{R} \left(\frac{1}{T_0} - \frac{1}{T} \right) \right] \quad (118)$$

where E_a is the activation energy and $T_0 = 300 \text{ K}$ is the reference temperature.

Oxygen fraction correction.

$$K(Y_{\text{O}_2}) = K_0 \frac{Y_{\text{O}_2}}{0.233} \quad (119)$$

EHD electric-field enhancement. An applied electric field modifies combustion via ionic-wind-enhanced transport (Lawton & Weinberg):

$$K_E = K_0 \left[1 + \alpha_E \left(\frac{E}{E_{\text{ref}}} \right)^2 \right] \quad (120)$$

with $\alpha_E \approx 0.05\text{--}0.3$ (fuel-dependent) and $E_{\text{ref}} = 10 \text{ kV/m}$.

15.4 Heat Release

The instantaneous heat release rate from a single particle is:

$$\dot{Q} = \left| \frac{dm}{dt} \right| \Delta H_c \quad (121)$$

The adiabatic flame temperature at stoichiometric conditions in air:

$$T_{\text{ad}} = T_0 + \frac{\Delta H_c}{c_{p,\text{mix}} (1 + \text{AF}_{\text{stoich}})} \quad (122)$$

15.5 Radiative Emission

The Stefan–Boltzmann radiative power from a spherical particle of diameter d at temperature T_p :

$$P_{\text{rad}} = \varepsilon_p \sigma \pi d^2 (T_p^4 - T_{\infty}^4) \quad (123)$$

The distinct flame colours observed in Figure Z.3 arise from element-specific electronic transitions. The peak emission wavelength follows Wien’s displacement law:

$$\lambda_{\text{peak}} = \frac{b}{T_p} \quad (b = 2.898 \times 10^{-3} \text{ m} \cdot \text{K}) \quad (124)$$

The spectral radiance at wavelength λ is given by Planck’s law:

$$B(\lambda, T) = \frac{2hc^2}{\lambda^5} \frac{1}{\exp \left(\frac{hc}{\lambda k_B T} \right) - 1} \quad (125)$$

15.6 Lagrangian Particle Phase

Each particle is tracked as a Lagrangian entity with state (x_p, v_p, d, T_p, m) . The equations of motion are:

Position.

$$\frac{dx_p}{dt} = v_p \quad (126)$$

Velocity (Stokes drag + EHD force).

$$\frac{dv_p}{dt} = \frac{u - v_p}{\tau_p} + \frac{F_E}{m_p} \quad (127)$$

where the particle relaxation time is:

$$\tau_p = \frac{\rho_p d^2}{18 \mu_g} \quad (128)$$

and $F_E = q_p E$ is the Lorentz force on a thermionically charged particle.

Temperature (heat balance).

$$m_p c_p \frac{dT_p}{dt} = \dot{Q}_{\text{gen}} - P_{\text{rad}} \quad (129)$$

Thermionic charge. Burning metal particles acquire charge via thermionic emission. The Rayleigh-limit estimate gives:

$$q_p \approx 4\pi\epsilon_0 r_p \frac{k_B T_p}{e} \quad (130)$$

15.7 Eulerian Source Terms (Two-Way Coupling)

Particle effects are projected onto the Eulerian grid as volumetric source terms. For N_p particles in a cell of volume V_{cell} :

Mass source.

$$S_{\text{mass}} = \frac{N_p}{V_{\text{cell}}} \left| \frac{dm}{dt} \right| \quad (131)$$

Energy source.

$$S_{\text{energy}} = \frac{N_p}{V_{\text{cell}}} \dot{Q} \quad (132)$$

Momentum source (drag coupling).

$$S_{\text{mom}} = \frac{18 \mu_g N_p}{\rho_p d^2 V_{\text{cell}}} (u - v_p) \quad (133)$$

Oxygen consumption.

$$S_{\text{O}_2} = -\frac{N_p}{V_{\text{cell}}} \left| \frac{dm}{dt} \right| \frac{0.233 \text{ AF}_{\text{stoich}}}{1 + \text{AF}_{\text{stoich}}} \quad (134)$$

15.8 Dimensionless Numbers

Damköhler number (first kind). The ratio of flow residence time to reaction burnout time:

$$\text{Da}_I = \frac{\tau_{\text{flow}}}{\tau_{\text{rxn}}} = \frac{L/U}{d_0^n/K} \quad (135)$$

$\text{Da} \gg 1$: fast reaction (mixing-limited); $\text{Da} \ll 1$: slow reaction (kinetics-limited).

Stokes number.

$$\text{St} = \frac{\tau_p}{\tau_f} = \frac{\rho_p d^2}{18 \mu_g} \cdot \frac{U}{L} \quad (136)$$

Mass loading ratio.

$$\phi_m = \frac{N_p m_p}{\rho_g Q} \quad (137)$$

$\phi_m < 0.1$: one-way coupling; $\phi_m > 0.1$: two-way coupling required.

Biot number.

$$\text{Bi} = \frac{h d}{2 k_p} \quad (138)$$

For metal particles $\text{Bi} \ll 1$ (lumped-capacitance valid).

Nusselt number (Ranz–Marshall).

$$\text{Nu} = 2 + 0.6 \text{Re}_p^{1/2} \text{Pr}^{1/3} \quad (139)$$

Particle Reynolds number.

$$\text{Re}_p = \frac{\rho_g |u - v_p| d}{\mu_g} \quad (140)$$

15.9 Fuel Comparison Framework

The five fuels are compared using:

- Gravimetric energy density: $e_g = \Delta H_c$ (J/kg)
- Volumetric energy density: $e_v = \rho \Delta H_c$ (J/m³)
- Burnout time for $d_0 = 10 \mu\text{m}$ reference particle
- Adiabatic flame temperature T_{ad}
- Emissivity and peak emission wavelength

Table 5: Fuel comparison for $d_0 = 10 \mu\text{m}$ particles

Fuel	e_g (MJ/kg)	e_v (GJ/m ³)	K (m ⁿ /s)	t_b (μs)	ε
CH ₄	50.0	0.033	—	—	0.10
Fe	7.4	58.3	3.0×10^{-7}	33	0.85
Al	31.1	84.0	2.0×10^{-6}	50	0.95
B/Al	40.0	100.0	1.5×10^{-6}	67	0.88
Zr	12.0	78.1	4.0×10^{-7}	68	0.80

Equation Index

This document contains numbered equations grouped across eight expanded phases and the preceding foundational sections. Equation numbering is sequential and cross-referenceable via `\eqref`.